

Foundation (Jalapeno)

- Q1.** What is the standard form of a quadratic function?
 (A) $ax^2 + bx + c$
 (B) $ax + bx^2 + c$
 (C) $ax^2 - bx + c$
 (D) $ax^2 + bx - c$
 (E) $ax + bx + c$
- Q2.** A basketball player scores $2x^2 + 3x + 5$ points in x games. What type of function represents the points scored?
 (A) Linear
 (B) Quadratic
 (C) Exponential
 (D) Polynomial
 (E) Rational
- Q3.** Identify the key feature of the quadratic function $f(x) = x^2 + 2x + 1$.
 (A) Vertex at $(0, 0)$
 (B) Vertex at $(-1, 0)$
 (C) Axis of symmetry at $x = -1$
 (D) Axis of symmetry at $x = 0$
 (E) Vertex at $(-1, -1)$
- Q4.** The distance d an athlete runs is given by $d = 2t^2 + 5t + 1$, where t is time. What is the type of function?
 (A) Quadratic
 (B) Linear
 (C) Exponential
 (D) Polynomial
 (E) Rational
- Q5.** What is the axis of symmetry for $f(x) = x^2 - 4x + 3$?
 (A) $x = 0$
 (B) $x = -2$
 (C) $x = 2$
 (D) $x = 4$
 (E) $x = -1$
- Q6.** A cyclist's distance traveled d is given by $d = 3t^2 - 2t + 1$. Identify the function type.
 (A) Quadratic
 (B) Linear
 (C) Exponential
 (D) Polynomial
 (E) Rational
- Q7.** What is the vertex form of a quadratic function?
 (A) $a(x - h)^2 + k$
 (B) $ax^2 + bx + c$
 (C) $ax + bx^2 + c$
 (D) $ax^2 - bx + c$
 (E) $ax + bx + c$
- Q8.** A marathon runner's pace is given by $p = 2t^2 - 5t + 1$. What is the type of function that represents pace?
 (A) Linear
 (B) Quadratic
 (C) Exponential
 (D) Polynomial
 (E) Rational

Open Ended Questions (FRQ)

- Q9.** Explain the difference between the standard form and vertex form of a quadratic function, providing an example of each.
- Q10.** Given $f(x) = x^2 - 4x + 3$, find the axis of symmetry and vertex of the parabola. Show your work and provide explanations.

Chef's Tips

- Tip 1: Encourage students to use the vertex form to identify key features of quadratic functions.
- Tip 2: Remind students that the axis of symmetry is given by $x = -\frac{b}{2a}$.
- Tip 3: Emphasize the importance of identifying the type of function in word problems.